

customer terminal capable of reaching speeds of 400 Mbps. In future, Amazon plans to become the closest rival to Starlink.

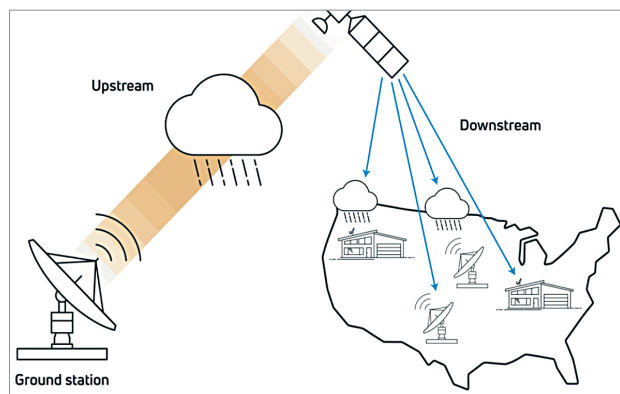
The European Union

The European Union (EU) approved a Euro 6 billion (\$6.11 billion) project in February 2022 to compete with Starlink to provide services in Europe. Airbus, SES, and Eutelsat have been commissioned to conduct a technical study for the project for the approval of the EU member states.

The Indian scenario

Currently, internet services are not available to nearly 75% of rural India due to lack of cellular or fibre connectivity. India, therefore, is a key emerging market providing \$1 billion near-term annual business opportunity (\$4.7 billion by 2025 as per E&Y estimates).

However, there are many challenges, such as the very high costs of developing the systems and launching of satellites—nearly \$20 (₹1,650) per GB, which is about ten times the global average. It is tedious to lease the foreign satellites capacity as the process involves obtaining several approvals, including a go-ahead from the Department of Space. The user access terminals are not available indigenously and the imported terminals are costly, taking the base model cost to nearly \$1,000.



Weather related disturbances are a bigger concern in case of satellite internet (Source: <https://news.viasat.com>)



SpaceX plans to deploy many more satellites in the future taking the total to 42,000 (Source: <https://www.mushroomnetworks.com>)



OneWeb is owned by India's Bharti Group and the UK government (Source: <https://telecomtalk.info>)

However, the private sector is constantly making efforts to launch the services and there is intense competition between Jio, Bharti-backed OneWeb, Nelco (a Tata group satcom company), Canada's Telesat, SpaceX, and Amazon. The Department of Telecom (DoT) has asked the Telecom Regulatory Authority of India (TRAI) to allocate 27.5-28.5GHz band, 71-76GHz and 81-86GHz, E band and 57-64GHz, and V band for satellite internet services.

Bharti Airtel's OneWeb

Hughes Communications India Private Ltd (HCIPL), a joint venture between Bharti Airtel's OneWeb

and Hughes, has received the national long distance (NLD) and global mobile personal communications by satellite (GMPCS) licences for delivering the services in India. The remaining regulatory approvals are expected to become available soon to commence

the pilot services. To provide the services in India, OneWeb plans to launch 36 satellites in cooperation with ISRO's NewSpace India Ltd (NSIL) from the Sriharikota centre.

Reliance Jio

Jio Satellite Communication Ltd (JSCL)

has a multi-year capacity purchase agreement with Luxembourg's SES, which provides satellite broadband services using geostationary and medium Earth orbit (MEO) satellite constellations. JSCL will be offering satellite broadband services to industries, aviation, and maritime users; and mobile backhaul (MBH) services to mobile networks to connect cell site air interfaces to wireline networks connected to data centres that host the content and applications accessed by mobile users, both human and machine. JSCL has received the licence for GMPCS services.

The Tata group

Nelco, a company of the Tata group, is partnering with Canadian satellite player Telesat to form a joint venture, Tata-Telesat, to launch the satellite broadband services, including the in-flight internet service on domestic and international flights under the brand 'Lightspeed' around 2024.

Amazon

Amazon has initiated efforts to bring high-speed satellite internet services to India using Project Kuiper satellite constellation and has approached the government to obtain the necessary regulatory approvals.

SpaceX's Starlink

The Indian launch of Starlink service, originally planned in 2022, remains uncertain as the DoT and the Telecom Regulator have asked the Indian arm